Trigonometric Identities

Concept

An identity is true wherever both sides are defined. Prove it by transforming one side.

Pythagorean

$\sin^2x+\cos^2x=1;\ 1+\tan^2x=\sec^2x;\ 1+\cot^2x=\csc^2x.$

Sum / difference

$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B.$

Cosine sum / difference

$\cos(A\pm B)=\cos A\cos B\mp\sin A\sin B.$

Double angle

$\sin2x=2\sin x\cos x;\ \cos2x=1-2\sin^2x=2\cos^2x-1.$

Tangent formula

$\tan(A\pm B)=(\tan A\pm\tan B)/(1\mp\tan A\tan B)$, where defined.

Example 1

Find $\sin75°$ exactly.

Solution

1. $75°=45°+30°$; apply the sine sum identity.
2. $\sin75°=(\sqrt2/2)(\sqrt3/2)+(\sqrt2/2)(1/2)$.
3. Result: $(\sqrt6+\sqrt2)/4$.

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Example 2

Prove $(1 - \cos^2 x)/\sin x = \sin x$ on its original domain.

Solution

1. Restriction: $\sin x \neq 0$.

2. Use $1 - \cos^2 x = \sin^2 x$.

3. Left side $= \sin^2 x/\sin x = \sin x$, as required.

Trigonometric Identities

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find $\cos 75°$ exactly.

2. Simplify $\sec^2 x - \tan^2 x$ on its domain.

3. Prove $(1 + \tan^2 x)\cos^2 x = 1$ on the original domain.

4. If $\cos x = 5/13$ and x is in quadrant IV, find $\sin 2x$.

5. Explain why $\sin(A + B)$ is not $\sin A + \sin B$.

Trigonometric Identities

Practice answers | check after trying

1. $\cos(45° + 30°) = (\sqrt{6} - \sqrt{2})/4$.

2. 1, with $\cos x \neq 0$.

3. Replace $1 + \tan^2 x$ by $\sec^2 x = 1/\cos^2 x$, then multiply; $\cos x \neq 0$ is required.

4. $\sin x = -12/13$, so $\sin 2x = 2(-12/13)(5/13) = -120/169$.

5. At $A = B = 30°$, the left is $\sin 60° = \sqrt{3}/2$, while the right is 1. A single counterexample disproves the proposed identity.